

FPM #2: Exploratory Data Visualization

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IV. Import & wrangle data, then create exploratory data viz

```
##~~~~~  
##                               Setup                               ----  
##~~~~~  
  
#.....load libraries.....  
library(tidyverse)  
library(here)  
library(lubridate)  
  
#.....import data.....  
  
calbats <- read_csv(here("data", "observations-675568.csv"))
```

```
##~~~~~  
##                               Data wrangling / cleaning                               ----  
##~~~~~  
  
# iNaturalist data quick check  
glimpse(calbats)  
summary(calbats)  
  
# check for issues  
colSums(is.na(calbats))  
  
# check quality grade column  
table(calbats$quality_grade) # 993 research grade  
  
# how many observations will be lost with cleaning for research grade only  
calbats %>%  
  filter(
```

```

    quality_grade == "research",
    captive_cultivated == FALSE,
    !is.na(latitude),
    !is.na(longitude),
    !is.na(observed_on)
  ) %>%
  nrow() # only miss 2

# simple clean data for exploration
bats_clean <- calbats %>%
  filter(
    captive_cultivated == FALSE,
    # remove NAs
    !is.na(latitude),
    !is.na(longitude),
    !is.na(observed_on)
  ) %>%
  mutate(
    date = ymd(observed_on),
    year = year(date),
    month = month(date, label = TRUE, abbr = FALSE)
  )

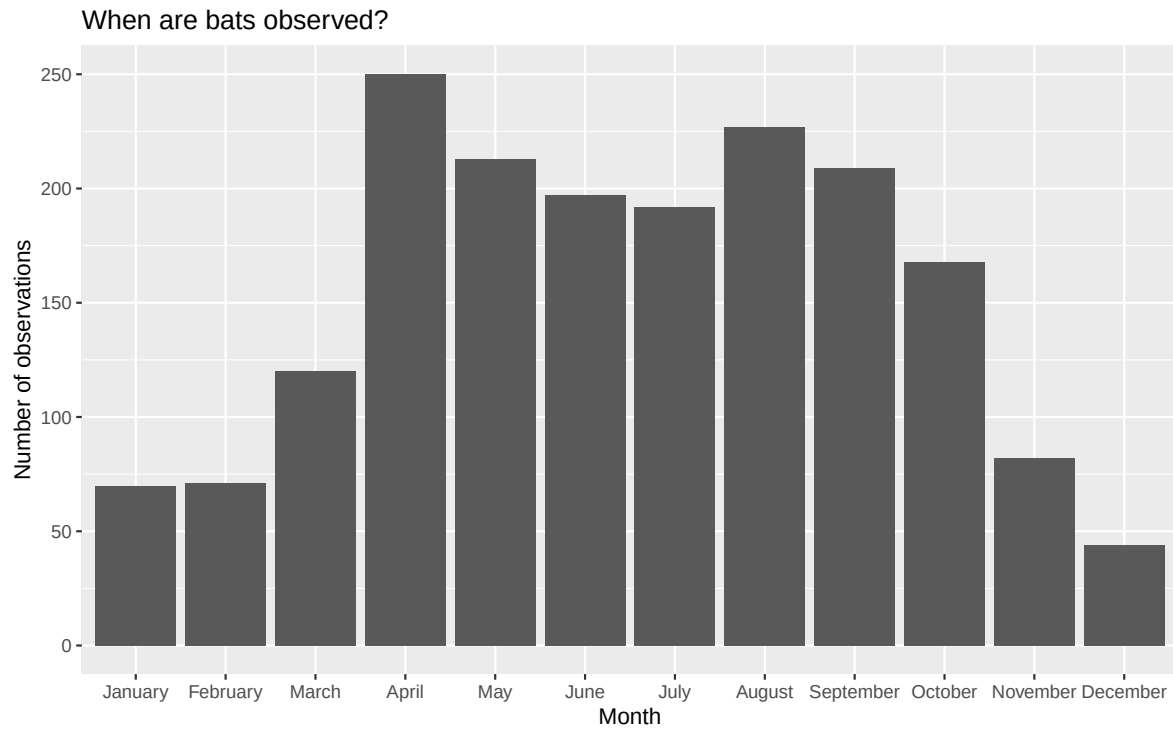
```

```

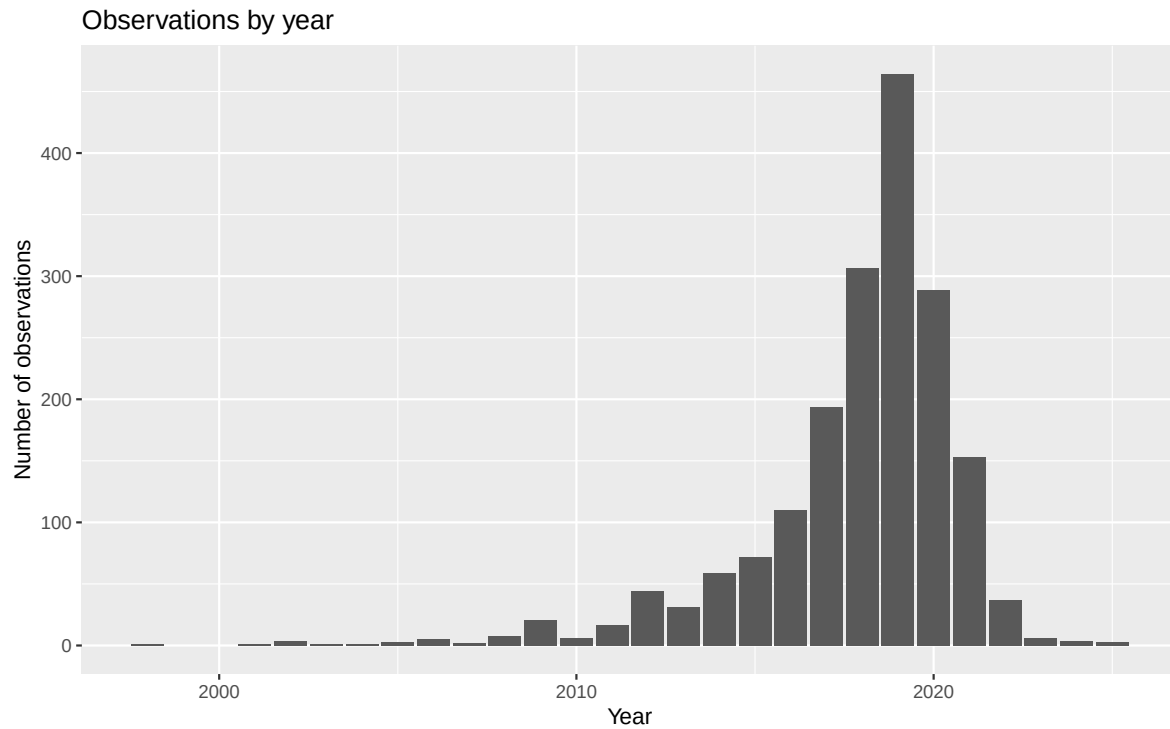
##~~~~~
##                                     Data visualization      ----
##~~~~~

# Plot 1: temporal patterns peak seasons/months
ggplot(bats_clean, aes(x = month)) +
  geom_bar() +
  labs(title = "When are bats observed?",
        x = "Month", y = "Number of observations")

```



```
# Plot 2: years with more observations
ggplot(bats_clean, aes(x = year)) +
  geom_bar() +
  labs(title = "Observations by year",
       x = "Year", y = "Number of observations")
```

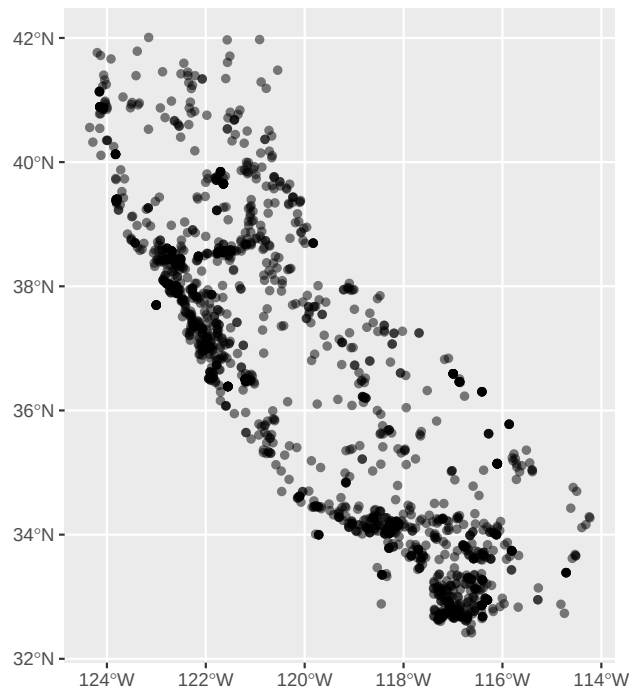


```
# spatial patterns
library(sf)

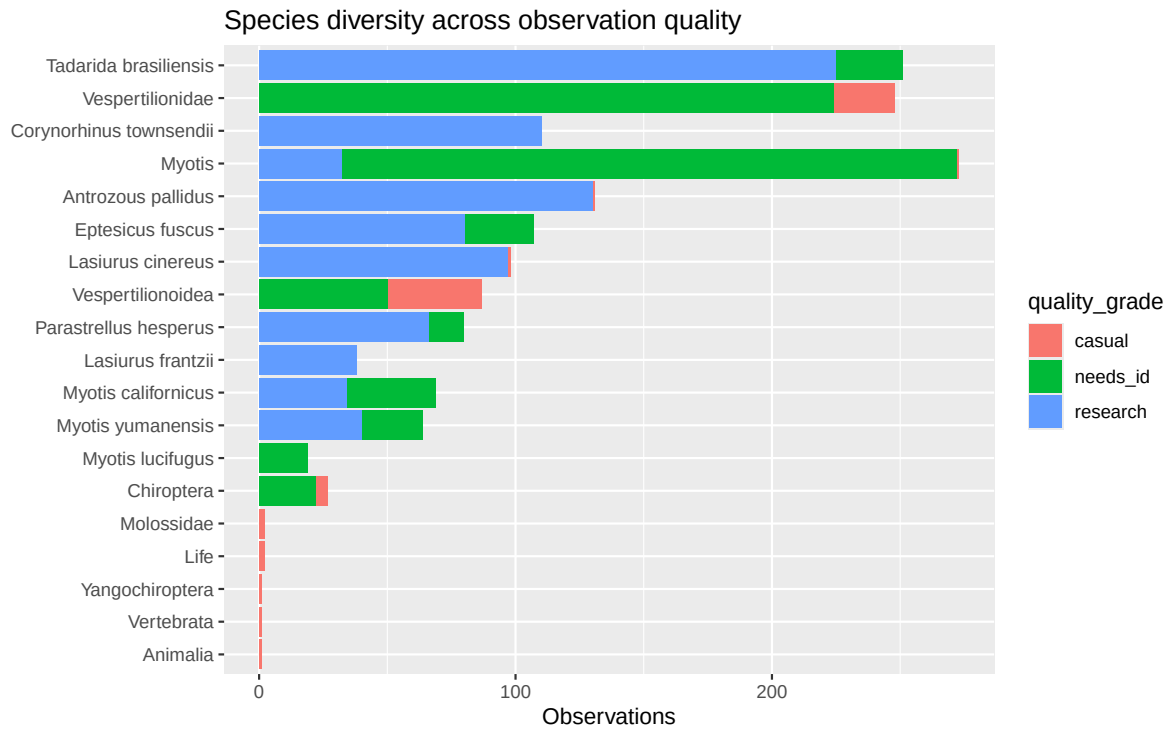
# Convert to spatial object
bats_sf <- st_as_sf(bats_clean, coords = c("longitude", "latitude"), crs = 4326)

# Plot 3
ggplot(bats_sf) +
  geom_sf(alpha = 0.5) +
  labs(title = "Where are bats being spotted?")
```

Where are bats being spotted?



```
# Plot 4: by quality grade to show citizen science engagement
bats_clean %>%
  count(scientific_name, quality_grade) %>%
  group_by(quality_grade) %>%
  slice_max(n, n = 10) %>%
  ggplot(aes(x = n, y = reorder(scientific_name, n), fill = quality_grade)) +
  geom_col() +
  labs(title = "Species diversity across observation quality",
       x = "Observations", y = NULL)
```



Check temporal and seasonal patterns as a table

```
table(bats_clean$year) # 2017-2020 sweet spot
```

1998	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
1	1	4	1	1	3	5	2	8	21	6	17	44	31	59	72
2016	2017	2018	2019	2020	2021	2022	2023	2024	2025						
110	194	307	464	289	153	37	6	4	3						

```
table(bats_clean$month) # April peak (149)
```

January	February	March	April	May	June	July	August
70	71	120	250	213	197	192	227
September	October	November	December				
209	168	82	44				

Creating an urban data to see if can focus on this topic

```
# Extract city patterns
bats_clean %>%
  filter(!is.na(place_guess)) %>%
  count(place_guess, sort = TRUE) %>%
  print(n = 30)

# filter to urban manually
bats_urban <- bats_clean %>%
  filter(str_detect(place_guess,
                    regex("Los Angeles|San Diego|San Francisco|Sacramento|Oakland|
                          San Jose|Santa Rosa|Berkeley|Fresno|Long Beach|
                          Riverside|Anaheim|Santa Cruz|Mill Valley",
                          ignore_case = TRUE)))
```

```
# Is there still enough species diversity?
bats_urban %>% count(scientific_name, sort = TRUE)
```

```
# A tibble: 29 x 2
  scientific_name      n
  <chr>             <int>
1 Tadarida brasiliensis 65
2 Vespertilionidae     27
3 Parastrellus hesperus 23
4 Lasiurus cinereus     21
5 Myotis               21
6 Vespertilionoidea    19
7 Myotis yumanensis    14
8 Lasiurus frantzii     13
9 Myotis californicus  13
10 Chiroptera           11
# i 19 more rows
```

```
# Is there still Enough temporal pattern?
table(bats_urban$month) # April still peaks
```

January	February	March	April	May	June	July	August
16	9	15	61	23	24	27	24

September	October	November	December
28	31	9	11

V. Answer questions

After completing the above steps, answer the following questions:

1. What have you learned about your data? Have any potentially interesting patterns emerged? Point to specific visualizations that you created as you describe your findings.

Plot 1: The month observation plot shows a very high peak in April and then peaks again in August. Something about early spring and late summer influences people to encounter/observe more bats. Plot 2: Observation peaked in 2017-2019, with sparse data before 2014 and a drop after 2020. I might decision to focus on observations from 2015 to 2025 since it has clearer distributions.

Plot 3: Shows that most observation are spread across the coast lines with dense clusters in Northern California and Southern California this make sense because large cities like San Francisco and Los Angeles have more people living in them.This helps to see if I could target my data to urban communities.

2. In FPM #1, you outlined some questions that you wanted to answer using these data. Have you made any strides towards answering those questions? If yes, how so? If no, what next steps do you need to take (e.g. I need to create X plot type, I still need to track down Y data, I need to restructure existing data so that you can visualize it in Z ways, etc.)? Have any new questions emerged?

My data comes from citizen science observations(1,843 observations). While bats are found statewide, based on my goal I want to focus on urban environments.I identified 278 observations across major metro areas.By doing some data wrangling and specifically filtering for big cites. I wont be able to answer ecological behaviors such as what they eat or what they are vulnerable to with this data but can try to include external facts about bat ecosystem services, conservation context, and actionable tips some how in my infographic by doing some research. What I could see with my data is what bats are present and in what areas. As well as when they are active and add a community engagement to my infographic New question I can ask focus more on what my data can provide. I can answer when they are active, is there diversity, and where there hotspots are in California or narrow to southern California.

3. What challenges do you foresee encountering with your data? These can be data wrangling and / or visualization challenges.

One challenge I can see is if I want to target urban observation how can I confirm the area is actual urban since most large cities have natural habitats such as Regional parks. iNaturalist's `place_guess` field is inconsistent, includes everything from specific addresses to value "Los

Angeles County” descriptions. Difficult to accurately identify truly urban observation without spatial analysis tools. Filtering to urban area focus story but risk sparse visualization. Not all observations are identified to species level. Deciding whether to include/exclude these affects species diversity counts. Data concentration in 2015-2020 required data filtering. Older data was too sparse and recent data is incomplete. Visualization challenges; how to balance species diversity with citizen science participation in limited space. Multiple species needs colorblind friendly palette that's distinct. Adding external context info that exists in the data.